Week 5 Update — COVID-19 Global Data Analysis

Shrey Jain • Correlation analysis: vaccination vs death rate • Prepared for Neelam Gupta

How to read this analysis. This is a **country-level (ecological)** correlation. It cannot establish individual cause and effect and is sensitive to confounding. The headline finding is a methodological one: the *naive* vaccination–death correlation is positive, but that is a confounding artifact (population age); after adjustment the association reverses to negative, consistent with vaccine benefit. Individual vaccine efficacy is established by clinical trials, not by this kind of analysis.

Weekly update

This week I tested the cross-country correlation between vaccination coverage and COVID death rate. To make the comparison fair, I related coverage *before* a wave to deaths *during* it (Omicron window, Dec 2021 – May 2022: vaccination as of 1 Dec 2021 vs deaths/million over the window), then used statsmodels to adjust for confounders.

Discovered:

- The **naive correlation is positive** (Pearson r ≈ +0.38; Spearman ≈ +0.64) — which taken literally would wrongly suggest vaccination raises deaths.
- It is **driven by confounding**: median age correlates with mortality far more strongly (r ≈ +0.70). Older, richer countries both vaccinated more and recorded more COVID deaths.
- **Adjusting for age and GDP flips the vaccination coefficient negative** — from about +130 to –**98 deaths/million per SD (p ≈ 0.003)**; partial correlation –0.24; model R^2 rises from 0.15 to 0.58.
- **Takeaway:** once the main confounder is accounted for, higher vaccination is associated with *lower* Omicron-period mortality — consistent with a protective effect, though ecological evidence supports rather than proves it.

Analysis

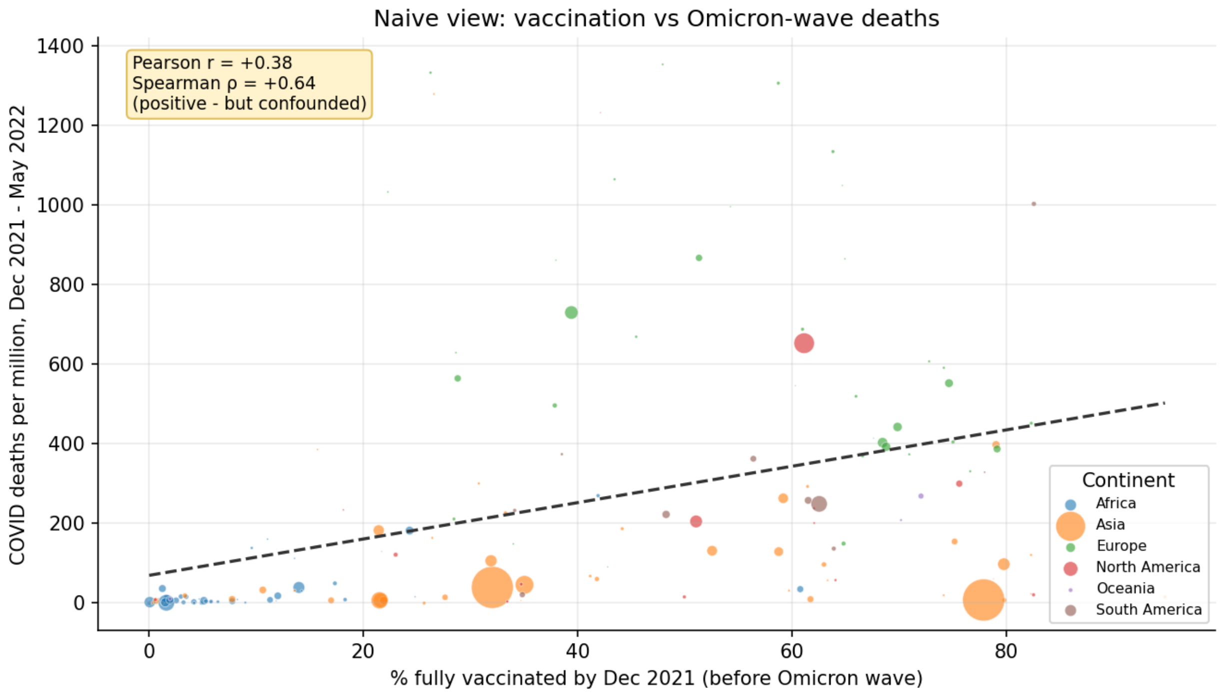

Fig 1. Naive view: vaccination (before Omicron) vs deaths during the wave. The upward slope is the confounded, misleading result.

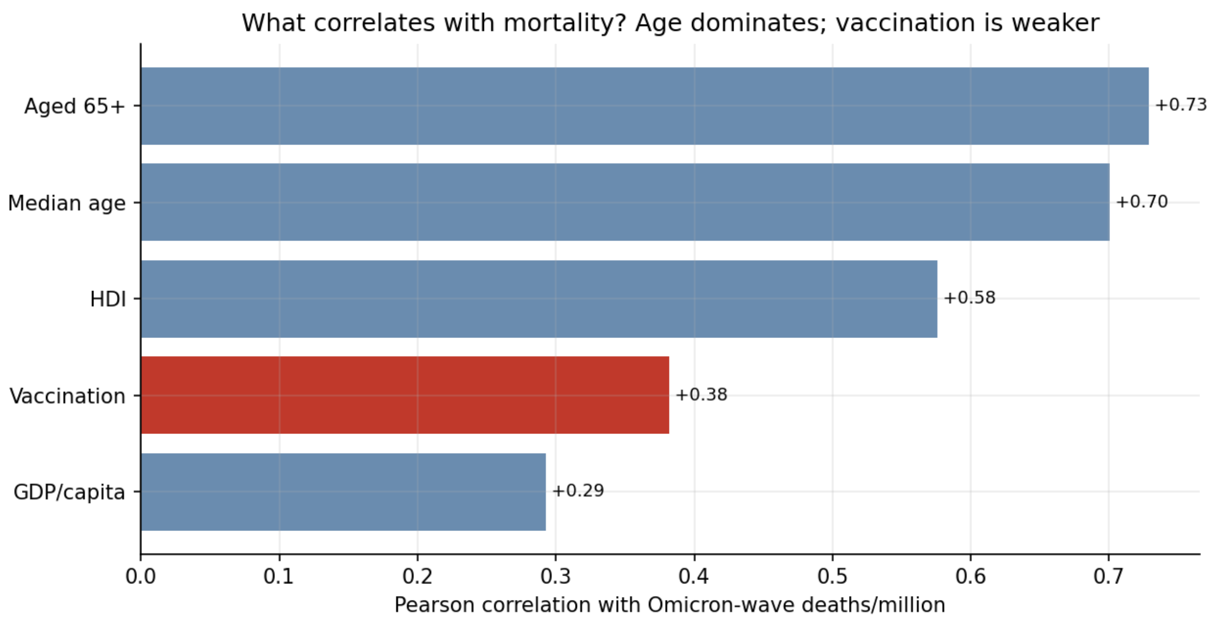

Fig 2. Median age and aged-65+ correlate with mortality far more strongly than vaccination does — the source of the confounding.

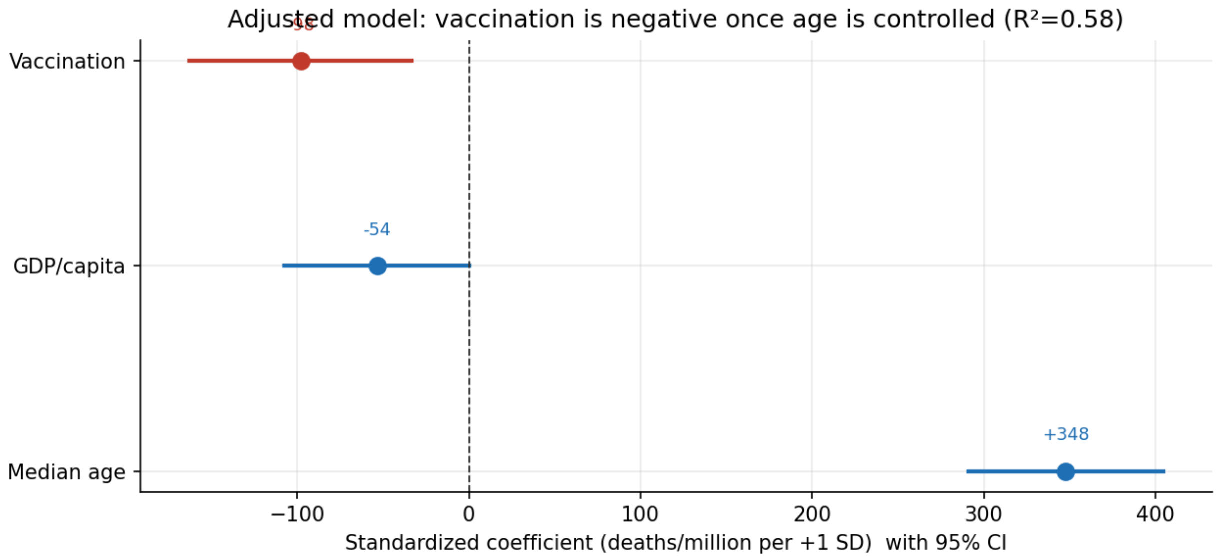

Fig 3. Adjusted regression (standardized): age is the dominant positive driver; vaccination is negative once age and GDP are held constant.

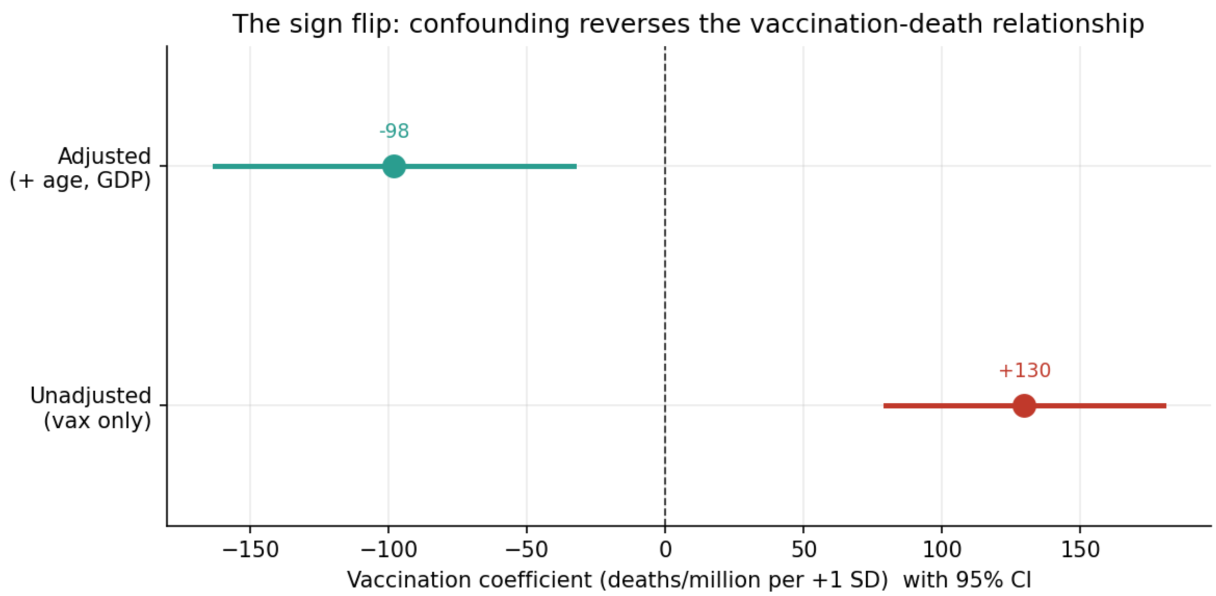

Fig 4. The punchline: the vaccination coefficient flips from positive (unadjusted) to negative (adjusted for age and GDP).

Work artifacts & data sources

Repository	github.com/shreyjain99/covid19-global-analysis
Week 5 folder	github.com/shreyjain99/covid19-global-analysis/tree/main/week%205
Week 5 notebook	week5_vax_vs_death_correlation.ipynb (Plotly + statsmodels)
Charts / data	figures/fig1–fig4 (.png), regression_summary.txt, vax_deaths_data.csv
Data source	Our World in Data COVID-19 dataset (github.com/owid/covid-19-data)

Reproducibility (how to re-run)

- `pip install -r requirements.txt`.
- Open `week5_vax_vs_death_correlation.ipynb` and Run All — it downloads the dataset and regenerates every figure and regression from the raw data.
- Or: `python scripts/analysis_week5.py` (charts + regression) then `python scripts/build_week5_report.py` (this PDF).

Risks / blockers & support needed

- **No blockers to delivery.** The week's objective is met and reproducible.
- **Interpretation risk (important):** the raw positive correlation is easy to misread. It is a confounding artifact, not evidence against vaccines; the report states this explicitly.
- **Method caveats:** ecological design (no individual inference), residual confounding (prior infection, healthcare capacity, reporting quality), country death-undercounting, and a single wave window as a simplification.
- **Support:** confirm whether the Omicron window and set of controls are acceptable, or whether a different wave / additional covariates are preferred.